

# 7.15 block 07b split receipt — 2026-09-26

## Decision and provenance

The owner authorized three stacked replacement review units after all three Copilot rounds on firmware PR #863 returned findings. PR #863 remains open as the historical complete candidate. Its final code head is 729ae7cfacf581ce534a98f6f5e0cc39efee3838, based on 8c654bcd8e1c8936a86aaa625a4ab78900a250e9. The reviews were 5323189342 (four findings), 5323944293 (two) and 5324788734 (one). All seven threads received source dispositions and are resolved. No fourth review is requested on the unsplit PR.

The #863 audit report and PDF were frozen before this authorization. Their "split pending" statement is a historical status; this receipt records the subsequent owner decision and replacement stack.

Replacement drafts: firmware [#865](https://github.com/BitHighlander/keepkey-firmware/pull/865) and [#866](https://github.com/BitHighlander/keepkey-firmware/pull/866), with companion Python [#114](https://github.com/BitHighlander/python-keepkey/pull/114) and [#115](https://github.com/BitHighlander/python-keepkey/pull/115).

| Unit                                               | Boundary | Local code head before this receipt      | Python companion                         |
|----------------------------------------------------|----------|------------------------------------------|------------------------------------------|
| A-C formatters and interpolated intent             | 7a base  | 75824620923c811e0c1e133879eb6d90ca63bdb6 | f813519c7c1163783b0e89b5cbd70536831f5e49 |
| D groups, one-array iteration, optional visibility | A-C      | 6cd843bf12e253d4a0cb32820869b2f8bdcdb74f | 8fc1c2bc038db8ee2c192b162c3308234117275e |
| E embedded calls and assembled audit               | D        | ddc47e32e3f19b94c2f3b8a7d022eca3a470522  | 67e891214086f7c7c6ebc7f75f9e3e9f27ec96a0 |

The E firmware source matches #863's tracked tree except for the Python gitlink and this receipt. The E companion's tracked tree matches #228's df376eec6ea7d705f7a70bdd31210f8cf929f97e; its new SHA gives it a reviewable diff against the D companion. Both equivalences were checked with `git diff --exit-code` before adding this receipt. Each unit adds only the source it compiles or dispatches; D and E source is absent from the A-C unit, and E source is absent from D.

## File placement

Each cell identifies the adjacent branch diff that owns that file's edits. For a shared file, edits introduced in A-C, D and E are distinct changes; later edits to an earlier line belong to the later diff. The three adjacent diffs account for the entire original #863 tracked change set. The Python gitlink identifies three companion branches with the same boundaries.

| Tracked path                                                   | A-C | D | E |
|----------------------------------------------------------------|-----|---|---|
| deps/python-keepkey                                            | ✓   | ✓ | ✓ |
| docs/release/audit-units/715-07b-formatters-audit-20260925.md  |     |   | ✓ |
| docs/release/audit-units/715-07b-formatters-audit-20260925.pdf |     |   | ✓ |
| docs/security/HANDOFF-ERC7730-715-FORMATTERS.md                | ✓   | ✓ | ✓ |
| docs/security/HANDOFF-ERC7730-PHASE-E.md                       |     |   | ✓ |
| include/keepkey/firmware/eip712_stream.h                       | ✓   |   |   |
| include/keepkey/firmware/erc7730_abi_stream.h                  | ✓   |   | ✓ |
| include/keepkey/firmware/erc7730_capabilities.h                | ✓   | ✓ | ✓ |
| include/keepkey/firmware/erc7730_catalog.h                     | ✓   | ✓ | ✓ |
| include/keepkey/firmware/erc7730_field.h                       | ✓   |   | ✓ |
| include/keepkey/firmware/erc7730_format.h                      | ✓   |   |   |
| include/keepkey/firmware/erc7730_program.h                     | ✓   |   |   |
| include/keepkey/firmware/erc7730_workflow.h                    | ✓   | ✓ | ✓ |

|                                              |   |   |   |
|----------------------------------------------|---|---|---|
| include/keepkey/firmware/ethereum.h          | ✓ |   |   |
| lib/firmware/CMakeLists.txt                  | ✓ |   |   |
| lib/firmware/eip712_stream.c                 | ✓ |   |   |
| lib/firmware/erc7730_abi_stream.c            | ✓ | ✓ | ✓ |
| lib/firmware/erc7730_capabilities.c          | ✓ | ✓ | ✓ |
| lib/firmware/erc7730_catalog.c               | ✓ | ✓ | ✓ |
| lib/firmware/erc7730_field.c                 | ✓ |   | ✓ |
| lib/firmware/erc7730_format.c                | ✓ |   |   |
| lib/firmware/erc7730_program.c               | ✓ |   |   |
| lib/firmware/erc7730_workflow.c              | ✓ | ✓ | ✓ |
| lib/firmware/ethereum.c                      | ✓ |   |   |
| lib/firmware/fsm.c                           |   |   | ✓ |
| lib/firmware/fsm_msg_ethereum.h              | ✓ | ✓ | ✓ |
| scripts/emulator/test_stack07_regressions.py | ✓ | ✓ | ✓ |
| unittests/firmware/CMakeLists.txt            | ✓ |   |   |
| unittests/firmware/erc7730_abi_stream.cpp    | ✓ |   |   |
| unittests/firmware/erc7730_catalog.cpp       | ✓ | ✓ | ✓ |
| unittests/firmware/erc7730_field.cpp         | ✓ |   | ✓ |
| unittests/firmware/erc7730_format.cpp        | ✓ |   |   |
| unittests/firmware/erc7730_program.cpp       | ✓ |   |   |
| unittests/firmware/erc7730_workflow.cpp      |   |   | ✓ |

## Finding placement and disposition

| Finding group                                                                                                                                                                                                                                      | Unit | Disposition                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|
| D1 numeric signer constants; D3 signer text and decimals; D4 native Wanchain state; D5 token table precedence; D6 intent titles; D7 line splitting; D8 control bytes and oversized raw values; D12 root signer path; D16 signer marks and raw unit | A-C  | Preload refusal or exact on-screen disclosure, with native and wire controls.       |
| D10 parallel array pairing; D18 fixed indices into dynamic arrays                                                                                                                                                                                  | D    | Bounds failures remain fail closed and are documented with wire cases.              |
| D2 embedded intent; D9 literal callee; D17 refused inner definition; inner @.value without amountPath; depth-two and call binding                                                                                                                  | E    | Refuse at preload or show blind, with inner-call and binding controls.              |
| D11 table, text and decimal limits                                                                                                                                                                                                                 | A-C  | The first companion mirrors preload bounds.                                         |
| D11 iteration path limits                                                                                                                                                                                                                          | D    | The second companion mirrors one-array iteration.                                   |
| D11 embedded-call limits                                                                                                                                                                                                                           | E    | The third companion mirrors inner-call capabilities.                                |
| D14 field title                                                                                                                                                                                                                                    | D    | Iteration fields use a numbered device-owned title.                                 |
| D15 pager continuation test handling                                                                                                                                                                                                               | E    | Runtime tests filter continuation pages without de-duplicating them.                |
| Copilot round 1: scalar formatter repeated in an iteration                                                                                                                                                                                         | D    | Value path must walk the active array; positive/negative firmware and mirror tests. |
| Copilot round 1: stale phase counts, E status and EX test count                                                                                                                                                                                    | E    | Handoffs and release report corrected.                                              |
| Copilot round 2: display frames overwrote path classes                                                                                                                                                                                             | D    | Dedicated display_frames, with a class-preservation test.                           |
| Copilot round 2: replay string reader accepted NUL                                                                                                                                                                                                 | A-C  | Independent reader rejects NUL, with a negative control.                            |

|                                                            |     |                                                                                      |
|------------------------------------------------------------|-----|--------------------------------------------------------------------------------------|
| Copilot round 3: signer native alias hid the token address | A-C | Signer alias and address precede the amount; maximum and Wanchain boundaries tested. |
|------------------------------------------------------------|-----|--------------------------------------------------------------------------------------|

The original 102-agent audit recorded 42 findings only as clusters D1-D19. The raw per-finding manifest was not retained. D13 and D19 were not named in the surviving handoff; this receipt cannot reconstruct or assign individual raw findings from those clusters. The 14 later re-audit defects are recorded in the #863 report as behaviors and controls, not as a raw manifest. These historical evidence gaps remain explicit; no finding is marked clean merely because its original row is unavailable. The seven Copilot comments are individually assigned above and resolved on #863.

Validation ledger

| Unit | Local native evidence                                                                                | Host/device compiler evidence                                          | Other gates                                                                    |
|------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| A-C  | firmware-unit 649/649 in a clean run directory; catalog, formatter, raw-overflow and replay controls | Two focused compiler/validator cases pass                              | ARM, full wire and hosted CI pending on split head.                            |
| D    | firmware-unit 652/652; iteration and display-frame controls pass                                     | Seven focused compiler/validator cases pass                            | ARM, full wire and hosted CI pending on split head.                            |
| E    | firmware-unit 656/656; tree-equivalent to #863 code                                                  | Compiler/catalog 23 pass, 2 skip locally without the official registry | #863 exact-head CI 36263598278 passed 16 required jobs; split-head CI pending. |

The canonical 7b pre-push script fails on inherited duplicate nanopb options, an absent report-gate test in the 7a base, and its own absent .cppcheck-build path. A corrected cppcheck invocation found no issues. The owner authorized the #863 push with this exception recorded. The split branches retain that same inherited preflight condition. Runs on all three split worktrees passed submodule cleanliness, unit-test registration and clang-format 20, then failed only those three inherited checks. Corrected cppcheck also completed without findings on A-C and D; E source is tree-equivalent to the prior corrected run on #863. Hosted CI and review results for replacement PRs belong to their exact heads and will be added to the PR descriptions.